

Hunger and Thirst

Food and drink are essential to human survival. Prompts by hormones to take in nutrients and water are experienced by the body as hunger and thirst.

Hunger

There are two types of hunger. Hedonic hunger involves eating food—particularly foods high in fat, sugar, and salt—when we are already full, while homeostatic hunger (see right) is a response to our energy stores depleting. Once food has passed through the stomach and intestines, the now-empty stomach releases a hormone called ghrelin. This acts on neurons in the hypothalamus to tell us that we are hungry, prompting us to eat. A hunger-inhibiting hormone called leptin is then released by adipose (fat-bearing) tissue to stop us from overeating.

Feeling hungry

The brain, digestive system, and fat stores form an interconnected system that regulates our feelings of hunger. The sensation of hunger can be caused by internal factors, such as our stomach being empty or our blood sugar levels falling, or by external triggers, such as seeing or smelling food.

DEHYDRATION AFFECTS OUR SHORT-TERM MEMORY, CONCENTRATION, AND ANXIETY LEVELS

5 Feeling full
Signals that leptin and insulin levels are increasing stimulate the hypothalamus to produce the hormone melanocortin, which makes us feel full.

4 Signals from adipose tissue
To prevent us from overeating, adipose tissue cells release a hunger-inhibiting hormone called leptin, which travels to the hypothalamus.

3 Signals from pancreas
After we have eaten, the small intestine releases the hormone incretin. This, combined with the stomach stretching and increased glucose in the blood, causes the pancreas to release insulin.

2 Urge to eat
Rising levels of ghrelin instruct the hypothalamus to release a chemical signal called neuropeptide Y, which stimulates our appetite.

Incretin produced by intestines triggers insulin production

Insulin levels tell hypothalamus whether body has enough energy

Decreased levels of leptin inform hypothalamus of low energy stores; increased leptin levels help inhibit appetite

Hypothalamus acts as regulator

Rising levels of ghrelin tell hypothalamus stomach is empty

Stretch receptors detect expansion of stomach

STOMACH

PANCREAS

Pancreas produces insulin

ADIPOSE (FAT) TISSUE

1 Empty stomach

Once the stomach has been empty for around two hours, levels of sugar and insulin in the blood decrease. This causes the stomach to produce the hormone ghrelin.

KEY

- Ghrelin
- Insulin
- Leptin
- Incretin
- Vagus nerve signal
- Movement of food